

RED DEER, AB, Canada

WMO: 718780

Lat: 52.179N

Lon: 113.893W

Elev: 2968

StdP: 13.19

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -24.5	(c) -17.8	(d) -33.0	(e) 0.9	(f) -24.1	(g) -26.6	(h) 1.4	(i) -17.6	(j) 23.3	(k) 20.3	(l) 20.3	(m) 16.8	(n) 4.4	(o) 200	(p) 0.671

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 22.7	(c) 82.3	(d) 63.0	(e) 79.0	(f) 61.4	(g) 75.9	(h) 60.1	(i) 65.8	(j) 78.0	(k) 63.7	(l) 75.5	(m) 61.7	(n) 72.9	(o) 8.2	(p) 160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.1	89.7	72.1	59.0	83.2	69.9	56.9	77.2	67.5	32.2	78.2	30.5	75.8	29.0	72.8	75.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 20.7	(b) 18.2	(c) 16.1	(e) -32.5	(f) 88.5	(g) 6.9	(h) 3.3	(i) -37.5	(j) 90.9	(k) -41.5	(l) 92.8	(m) -45.4	(n) 94.6	(o) -50.4	(p) 97.0		
(4) 20.7	(5) 18.2	(6) 16.1	(7) -32.7	(8) 69.1	(9) 6.8	(10) 2.5	(11) -37.6	(12) 70.9	(13) -41.6	(14) 72.4	(15) -45.5	(16) 73.8	(17) -50.4	(18) 75.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.0	11.6	14.0	24.1	38.3	49.5	56.7	61.4	59.4	51.0	38.9	24.0	13.3		
	DBStd	20.86	15.08	13.44	13.93	9.18	7.86	5.35	5.19	5.72	7.53	8.42	12.48	13.67		
	HDD50	5827	1192	1008	804	357	105	8	1	4	76	353	781	1138		
	HDD65	10272	1657	1428	1269	800	480	253	135	186	422	809	1230	1603		
	CDD50	1071	0	0	0	8	90	209	355	295	105	8	1	0		
	CDD65	44	0	0	0	0	1	5	24	13	1	0	0	0		
	CDH74	1084	0	0	0	3	76	109	405	366	122	3	0	0		
	CDH80	208	0	0	0	0	8	13	85	80	22	0	0	0		
	WSAvg	6.8	6.2	6.1	6.9	8.2	8.4	7.5	6.4	5.9	6.5	7.2	6.5	6.3		
	PrecAvg	19.9	0.9	0.7	0.9	1.1	2.4	3.9	3.4	2.6	1.6	0.8	0.8	0.8		
(15) Precipitation	PrecMax	24.1	2.4	1.7	1.9	3.4	5.0	7.5	8.4	4.6	3.1	2.0	2.4	1.9		
	PrecMin	14.2	0.2	0.2	0.1	0.2	1.0	0.8	0.9	0.6	0.2	0.3	0.1	0.2		
	PrecStd	2.4	0.5	0.3	0.4	0.7	0.9	1.3	1.5	1.1	0.9	0.5	0.6	0.4		
	PrecStd	2.4	0.5	0.3	0.4	0.7	0.9	1.3	1.5	1.1	0.9	0.5	0.6	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.6	44.3	56.4	72.0	80.8	81.4	86.6	86.8	83.0	72.3	58.5	46.0		
		MCWB	36.8	36.4	43.8	50.9	56.3	61.8	65.9	64.1	59.9	52.5	45.6	36.9		
	2%	DB	40.5	39.8	49.3	65.9	75.7	77.0	82.3	81.8	77.4	64.5	50.0	40.8		
		MCWB	34.6	33.5	39.5	48.6	54.6	59.4	65.3	62.7	58.3	49.6	40.7	34.6		
	5%	DB	36.4	36.2	44.8	60.3	71.1	73.5	78.5	78.1	72.8	58.8	44.1	36.4		
		MCWB	31.6	31.0	37.0	45.0	52.9	57.8	63.6	61.7	56.3	46.5	36.8	31.2		
	10%	DB	32.0	32.6	41.0	54.8	66.6	70.0	75.0	74.3	67.5	53.8	39.9	32.4		
		MCWB	28.2	28.6	34.7	42.2	50.5	56.2	62.1	60.5	54.1	43.9	34.3	28.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.3	37.0	44.6	51.9	58.3	64.6	69.5	67.3	62.1	54.1	46.0	37.8		
		MCDB	43.4	43.2	56.1	70.1	75.5	77.3	81.6	80.2	77.3	69.3	58.3	44.4		
	2%	WB	34.8	33.9	40.3	48.9	55.8	61.6	66.9	64.9	59.2	50.1	41.2	34.7		
		MCDB	40.1	39.3	48.3	64.6	72.2	73.1	79.0	77.5	74.4	63.3	49.0	40.3		
	5%	WB	31.9	31.3	37.3	45.9	53.7	59.3	64.8	62.9	57.0	47.2	37.5	31.6		
		MCDB	36.1	35.7	44.5	58.7	69.4	70.7	76.6	75.3	71.4	57.5	43.2	36.1		
	10%	WB	28.2	28.6	34.8	42.9	51.5	57.3	62.8	60.9	54.6	44.4	34.6	28.5		
		MCDB	31.9	32.6	40.5	53.5	64.8	68.0	73.2	73.2	66.9	52.8	39.3	32.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.2	20.7	19.2	21.7	23.4	20.7	22.7	24.2	24.6	22.9	18.9	18.8		
		MCDBR	22.5	24.0	24.0	33.7	32.3	28.2	29.2	32.7	35.8	32.4	25.1	22.0		
		MCWBR	17.8	19.0	16.2	18.8	15.7	13.9	15.2	16.5	18.6	19.5	17.3	16.9		
	5% WB	MCDBR	22.0	23.3	23.5	31.3	29.7	24.5	27.2	29.1	32.9	31.0	23.6	21.9		
		MCWBR	17.5	18.7	16.4	18.1	14.7	13.0	15.0	15.6	17.4	19.0	16.8	17.2		
(40) Clear-Sky Solar Irradiance	taub		0.261	0.267	0.293	0.328	0.338	0.345	0.346	0.359	0.309	0.282	0.253	0.244		
	taud		2.330	2.406	2.420	2.363	2.390	2.398	2.418	2.365	2.477	2.537	2.476	2.330		
	Ebn at Noon		240	276	289	288	290	288	285	275	279	268	247	227		
	Edh at Noon		19	23	29	35	35	36	34	34	27	21	16	16		
(44) All-Sky Solar Radiation	RadAvg		339	655	1093	1480	1777	1884	2001	1643	1149	692	362	257		
	RadStd		26	50	77	126	120	128	112	112	125	77	31	22		

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (74 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page